



Swing Trader · After-Fee Workflow

One question, asked the same way at every stage. **Is there a rule the model can trade that clears the fee, on data it has never seen?** Nothing is kept without passing that test, and no number is printed here that was not read out of the file that produced it.

WORKFLOW CHEAT SHEET

seamusrobertmurphy/swing-trader
Crypto NO-GO · Equities live on paper
State of 7 September 2026

1 Where the numbers come from

TWO ARCHIVES

Two filing cabinets, not two live feeds. Both were downloaded once and are read off disk, so a result rebuilt next year comes out the same.

Crypto. Binance's own monthly files of open, high, low, close and volume, one zip per coin per month, each checked against the exchange's published checksum.

Shares. Alpaca daily bars, adjusted for splits and dividends. On 26 August 2026 the screen read 12,571 live US companies and kept the **2,660** trading over \$20m a day above \$3 a share.

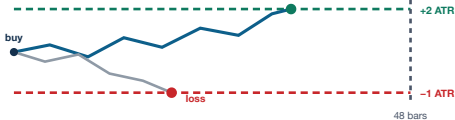
PANEL	ROWS	NAMES	INPUTS
5 min	5.53M	11	96
1 hour	1.64M	47	61
4 hour	3.94M	567	90
1 day	0.52M	540	104
shares 1d	5.00M	2,547	96

DEAD COINS ARE KEPT The archive lists **612** coin pairs that ever traded against **487** alive on 6 September 2026. Build the list from today's exchange and you quietly delete every coin that failed, which makes any result on it an upper bound. The 1-hour panel above is still the old 47-coin build.

5 What counts as a win

THREE BARRIERS

A win is a decision someone makes, not a fact waiting to be found. Three lines are drawn from the moment of buying: a profit to aim at, a loss to bail out at, and a deadline. Whichever the price touches first is the answer.



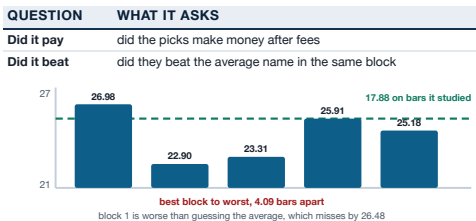
The two lines are set in units of the coin's own daily travel, so a calm coin and a violent one are judged on equal terms instead of on the same percentage.

A TARGET CAN LOSE BEFORE THE MODEL SPEAKS On the 4-hour panel the profit line is reached first on **254** bars in every thousand. Aiming twice as far up as down, you need **333** to come out level. The arithmetic is losing before anything is fitted, so the first thing to check is always the target's own record, not the model's.

9 Passing the bar

FOLDS, NOT TOTALS

A grand total can be carried by one lucky year. So history is cut into half-year blocks and the edge has to show up in at least **60 per cent** of them, on two questions at once.



NEVER READ THE AVERAGE ALONE Those five blocks average out to exactly what a steady 24.9 would, and a steady 24.9 would be a usable model. These are not.

2 Getting into the universe

ALL FOUR OR OUT

Four doors, and a coin has to pass all four before it is even considered. The test is re-run on every bar using only what was known on that day, because a filter applied with hindsight is not a filter.

DOOR	BAR IT HAS TO CLEAR
Trading enough	≥ 30m USDT in 24h
Moving enough	ATR 2.5–12% of price
Cheap to cross	≤ 0.5% estimated
Old enough	≥ 120 days lived

ATR, the average true range, is how far the price usually travels in a day, written as a share of the price. The band above is the daily one; on a shorter bar it shrinks with the square root of the bars per day, so a 4-hour bar wants 1.0 to 4.9%.

The **spread** is the gap between what buyers offer and sellers ask, and it is paid on the way in and again on the way out. The archive holds no order book, so it is estimated from each day's high and low by the Corwin and Schultz method and gated at 0.5%. The live screen reads the real spread and gates at 0.05%.

RANK WIDE, BUY NARROW The whole market is scored, then only coins that actually carried the edge on unseen data are allowed through at the moment of buying.

6 Splitting time

NEVER SHUFFLED

Shuffle the rows and you put next Tuesday in the study pile, then test on last Monday. Prices carry over from one bar to the next, so the future leak the future leak backwards and every score comes out flattering.



The yellow sliver is the **gap**. Without it the last rows studied are still waiting on outcomes that land inside the test window, the same leak wearing a different hat.

ONCE MEANS ONCE The final year is opened at the end and scored a single time. A setting chosen by looking at it has spent it, and the year cannot be made unseen again.

AS CONFIGURED

PIECE	SETTING
Held back	the final 365 days
Studied	all history before it
Gap at the cut	8 days (48 4h bars)

10 Rank, do not predict

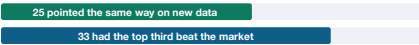
THE SURVIVING IDEA

Two different questions, and only one of them has ever produced anything. **Will this coin go up?** failed on every frame. **Which of these coins is strongest right now?** did not.

EVERY BAR, SORT THE WHOLE MARKET



Of 42 ways of sorting tried, 24 June 2026:



And still losing, because the market itself lost 0.382% a trade.

TWO CLAIMS, KEPT APART The sorting works and has never been disproved. The one way of *trading* it that was built, waiting for bitcoin to trend up first, held in 27 per cent of blocks against the 60 the rule wants, and was killed. Only the second sentence is settled. Untried: holding longer to pay the toll less often, and running the same sort on shares, where the market drifts up.

3 What the model reads

FOUR GROUPS

THE COIN'S OWN PRICE HISTORY		
day-length windows	short windows	chart indicators
WHICH WAY THE TREND POINTS		
Supertrend	adaptive version	calm or wild
WHAT THE REST OF THE MARKET IS DOING		
bitcoin, leading	slower bars above	crowding
WHO WAS IN A HURRY TO TRADE		
buyers against sellers	order-book texture	

Every input reads only bars that have already closed, and every one is a ratio, so it means the same on a \$90,000 coin as on a coin worth a tenth of a cent.

TWENTY-FOUR NUMBERS DO THE WORK OF NINETY Asked how many bars a trend has left, the 90 built inputs miss by **24.213** bars on the 4-hour panel. Twenty-four plain past returns, nothing built at all, miss by **24.247**. Guessing the average misses by 26.477. The two real answers sit a thirty-fourth of a bar apart, so they hold the same information and a memory-based model has no gap left to close.

7 How a model is judged

TWO DEFECTS

Every model answers with a probability, not a yes, so the score is how far that probability sat from what happened. Saying eighty per cent and being wrong costs more than saying fifty-five.

RMSE average miss, with big misses squared first, so one disaster moves it **ratio** the miss on unseen data divided by the miss on studied data

MODEL	STUDIED	UNSEEN	RATIO
HistGBM	.2619	.4818	1.840
LightGBM	.2806	.4826	1.720
Random forest	.4526	.4840	1.069
Logistic	.4855	.4909	1.011

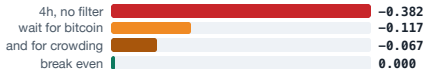
1-hour panel, 6 September 2026. A ratio near 1 means it does about as well on new data as on old. Above 1.1 it has memorised, and is thrown out.

TWO FAULTS, BOTH IN THE JUDGING The winner is chosen on the unseen miss alone and never consults the ratio, so it crowned HistGBM at **1.840** where the rule rejects past 1.1. And every model is trained to treat rare wins as though they were as common as losses, pushing its probabilities away from the very thing this score measures: on 4-hour bars it misses by **0.4844** where flatly answering "25.4 per cent, always" misses by **0.4353**. Nothing yet checks whether a stated 30 per cent happens 30 per cent of the time.

11 Where it stands

THE SCOREBOARD

Kept per crypto trade after fees, on bars never studied. A year of work moved the line and never crossed zero.



MOST LOSE BEFORE THE FEE Fee stripped out entirely: daily bars **-1.150%** a trade, 4-hour **-0.182%**, on 7 September. The 5-minute frame is the one exception, ahead by **+0.033%**, which the 0.20% toll wipes out six times over.

THE ONE THAT WORKS

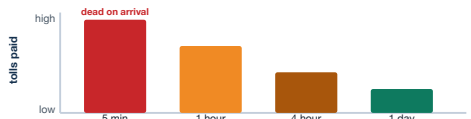
Twelve-month share momentum. Buy what rose most over the past year, skipping the latest month, and hold. Over 116 months it beat the average share by **1.006** percentage points a month, a gap **2.41** times its own standard error, what it would wobble on a rerun; past about two is the bar for calling a result real. It paid in 85 per cent of blocks and beat the market in 80.

Closed, with their numbers kept: 1-hour direction, a coin flip; sharper entries, 29% of blocks; the trend and breadth gate, 27%; 5-minute scalping, the fee wall.

4 The fee wall

THE WHOLE STORY

Every trade pays a toll going in and another coming out. A faster clock pays it more often, which is why the 5-minute idea died before any model was asked.



the argument in one line
kept = move - round trip - slippage
allow if kept ≥ 1.5%

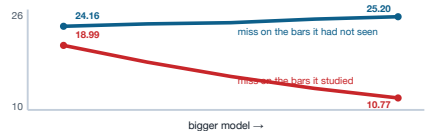
0.20 % assumed round trip	0.15 % best case round trip	1.5 % the least worth having
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THIS IS THE WALL. A basis point is a hundredth of one per cent. The best crypto edge measured here is five to twelve basis points *smaller* than the fifteen to twenty it costs to open and close the trade. No door, target, clock speed or new input closed that gap.

8 What tuning showed

SMALLER, NOT BIGGER

Nine settings, each tried on three separate stretches of the 4-hour panel, all asked the same question: how many bars until the trend turns?

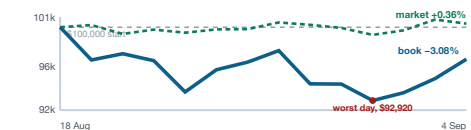


The miss on studied bars nearly halves while the miss on unseen bars gets worse. All nine settings fail the 1.1 rule, and the one that came top was the least powerful in the grid.

WHEN BEATS HOW, FORTY-ONE TO ONE Three quite different ways of feeding the model ended up **0.094** bars apart. The same model asked about a calm stretch of the market rather than a rough one moved **3.83** bars. Which months you test on matters forty-one times more than how you build the thing.

12 The live book

PAPER, REAL PRICES



The account's own daily closes, all fourteen, checked against the holdings valued at Friday's real closes: \$96,916.58 from \$100,000. The plan expects an 8.2% swing in that time, so a 3.1% dip is four tenths of an ordinary move.

49 shares held at 1.8% each	3/6 clean weekly cycles done	7.6 bp a fill mid-session
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weekly, never at the opening bell
alpaca_trade.py rebalance --execute

THIS IS FAKE MONEY A paper account at real prices; every order path stops unless it reads the exact word *true*. Fixed in code, not assumed from the venue: never short, never borrow, never options, never buy more of a loser, 5% most in one name, 10% least in cash.

TRAPS THAT COST DAYS

every one of these happened here, and each was found late

A number quoted from memory

Three figures on the previous version of this sheet were wrong because they were recalled rather than read. Open the file that produced the number, every time, including your own.

Rendering is not checking

A page can build, exit cleanly, and report 0 of 6 cycles where there are 3. Compare the new render against the last one before believing it.

The same error, twice

The overfit ratio was written upside down, corrected on 7 September in one file, and found the same day still inverted in a second. A model reported as stable at 0.902 was really 1.109, which the rule rejects.

A paper account is not a mark

On 7 September, a market holiday, the broker reported \$98,362 while the holdings at Friday's real closes came to \$96,917. Value the positions yourself before reporting a number.